

Macroinvertebrate and Fish Methods for the Lower Boise Biological Condition Gradient (BCG) Models (5/25/2026)

Users can run any samples through the BCG calculator and get a result. However, if samples were not collected and processed using the specific field and laboratory protocols below, results should be interpreted with caution because they are outside the experience of the model.

Macroinvertebrates

USGS National Water-Quality Assessment (NAWQA) Richest Targeted Habitat (RTH) method, which is a standardized, semi-quantitative method that targets the instream habitat expected to support the highest diversity and abundance of macroinvertebrates (typically riffles with coarse substrate). Collect five samples (0.25 m² each) for a total area of 1.25 m². Sampling period varies; the most recent samples were collected during the fall (September-November). The subsampling target count is 500 organisms. Identify Chironomidae (midges) to genus/species group, worms to class Oligochaeta, mites to subclass Acari, and the rest to the lowest practical level of resolution (typically species or genus).

Fish

USGS National Water Quality Assessment (NAWQA) Program (Meador and others, 1993) or U.S. Environmental Protection Agency's Environmental Monitoring and Assessment Program (EMAP) fish sampling protocols (Peck and others, 2002). Typically consists of two electrofishing passes: one through deep-water areas with a boat or tow barge; the second through shallower riffle areas with a backpack electrofisher to capture small benthic species often missed by boat electrofishing. Sample 40 times the mean channel width or about 1 mile. Netting crews consist of four to six people. Sampling occurs during low-flow periods, usually October-December. Identify all fish observed to the lowest practical level of resolution (typically species), and length information for the species listed below.

Age class assignments are based on Table D3 in the 2024 City of Boise report: §316(a) Demonstration Project for Boise Water Renewal Facilities. Table D3 is based on Wydoski and Whitney (2003)¹ who calculated age according to the species-specific relationships of age and growth, reporting the average total length at the end of each year of life for each species. Species age classes vary. Several species have evolved to be longlived and mature at 3 to 6 years such as northern pikeminnow, suckers, and mountain whitefish. Trout, sculpin, and dace have shorter life spans but also reach sexual maturity earlier at 2 to 3 years.

¹ Wydoski, R.S. and R.R. Whitney. 2003. Inland Fishes of Washington, Second Edition. Copublished with American Fisheries Society. Publisher University of Washington Press.
<https://uwapress.uw.edu/book/9780295983387/inland-fishes-of-washington/>

Age classifications were calculated from total measured length for the Representative Important Species (RIS) salmonid species (i.e., rainbow trout, brown trout, and mountain whitefish), catostomid species (bridgelip, mountain, and largescale sucker), cottid species (Columbia and shorthead sculpin), cyprinid species (longnose, riffle, leopard, and speckled dace; northern pikeminnow, and redbelt shiner) and centrarchid species (smallmouth and largemouth bass).

Table 1. Representative Important Species (RIS) age class and species length in millimeters (mm) as provided in Wydoski and Whitney (2003).

TaxaName	CommonName	AgeClass	Min Length (mm)	Max Length (mm)
CATOSTOMUS COLUMBIANUS	BRIDGELIP SUCKER	1	0	64
		2	65	147
		3	148	224
		4	225	292
		5	293	335
		6	336	361
		7	362	386
		8	387	401
		9	402	409
		>9	410	1500
CATOSTOMUS MACROCHEILUS	LARGESCALE SUCKER	1	0	46
		2	47	84
		3	85	132
		4	133	185
		5	186	325
		6	326	396
		7	397	422
		>7	423	1500
COTTUS CONFUSUS	SHORTHEAD SCULPIN	1	0	53
		2	54	83
		3	84	109
		4	110	119
		5	120	130
		>5	131	1500
COTTUS HUBBSI	COLUMBIA SCULPIN	1	0	53
		2	54	83
		3	84	109
		4	110	119
		5	120	130

		>5	131	1500
MICROPTERUS DOLOMIEU	SMALLMOUTH BASS	1	0	125
		2	126	224
		3	225	313
		4	314	382
		5	383	447
		>5	448	1500
MICROPTERUS SALMOIDES	LARGEMOUTH BASS	1	0	75
		2	76	149
		3	150	216
		4	217	269
		5	270	315
		>5	316	1500
ONCORHYNCHUS MYKISS	RAINBOW TROUT	1	0	94
		2	95	147
		3	148	175
		4	176	220
		5	221	306
		>5	307	1500
PANTOSTEUS PLATYRHYNCHUS	MOUNTAIN SUCKER	1	0	31
		2	32	64
		3	65	97
		4	98	127
		5	128	150
		6	151	170
		7	171	188
		8	189	206
		9	207	221
		>9	222	1500
PROSOPIUM WILLIAMSONI	MOUNTAIN WHITEFISH	1	0	113
		2	114	174
		3	175	220
		4	221	254
		5	255	290
		6	291	323
		7	324	352
		8	353	374
		>8	375	1500
PTYCHOCEILUS OREGONENSIS	NORTHERN PIKEMINNOW	1	0	73
		2	74	168
		3	169	262

		4	263	317
		5	318	377
		6	378	423
		7	424	464
		8	465	513
		9	514	556
		10	557	575
		>10	576	1500
RHINICHTHYS CATARACTAE	LONGNOSE DACE	1	0	30
		2	31	64
		3	65	97
		4	98	127
		5	128	150
		>5	151	1500
RICHARDSONIUS BALTEATUS BALTEATUS	COLUMBIA REDSIDE SHINER	1	0	53
		2	54	86
		3	87	119
		4	120	142
		5	143	155
		>5	156	1500
SALMO TRUTTA	BROWN TROUT	1	0	81
		2	82	137
		3	138	193
		4	194	244
		5	245	345
		>5	346	1500